

AI MEAL COACH : A NUTRITION/DIET RECOMMENDATION SYSTEM

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Abstract- Rapid urbanization and our fast-paced lifestyles have led to a notable rise in diet-related health issues like obesity, diabetes, and malnutrition. Many manual calorie-tracking apps require constant effort from users and can often be inaccurate. On the other hand, current nutrition platforms usually lack the kind of personal touch that considers an individual's unique health needs. To address this problem we have developed AI Meal Coach - A application that uses computer vision, machine learning, and Conversational AI, all in one place. It can be used for estimate nutrition and provide personalized diet recommendations. At its core, AI Meal Coach uses a custom-trained YOLOv8 model for real-time food detection from images users upload, with a focus on Indian cuisine [5]. It also incorporates a detailed nutrition lookup method that utilizes both a local food database and the USDA FoodData Central API [14]. Furthermore, we created a three-part machine learning approach that includes an intent classifier based on TF-IDF, a Random Forest model that generates personalized fitness roadmaps [13], and a rule-based meal planner that ensures safety in meal recommendations. We made our system user friendly so that it helps people get started and have conversations with it. Our tests show that the food detection works well the suggestions are always the same and the system is easy to use. When we looked at nutrition websites we saw that our system is better because it knows about a lot of Indian foods it uses many artificial intelligence models and it makes sure the dietary suggestions are safe. Our system stands out because of these things the food detection and the suggestions, which are reliable and consistent. The system is also very good, at recommending food. It does a great job of helping people make safe dietary choices with the food detection and the recommendations.

Keywords— Nutrition Estimation, Food Detection, Conversational AI, Computer Vision, Indian Cuisine, Personalized Health.

I. INTRODUCTION

Chronic lifestyle diseases like obesity, type 2 diabetes and malnutrition are becoming more common. That has pushed

preventive healthcare to the forefront especially when it comes to monitoring what people eat [4], [10]. The situation gets trickier in India because there is a huge diversity in cuisines and traditional foods are barely found in mainstream nutrition databases or health apps. Even though mobile health platforms are everywhere most of these platforms ask users to enter what they eat they depend on generic calorie charts and they deliver recommendations that do not fit the individual needs of the people who use them. Deep learning-based food recognition systems have begun to address this gap in lifestyle diseases like obesity, type 2 diabetes and malnutrition yet the integration of these systems with personalized dietary guidance for chronic lifestyle diseases, like obesity, type 2 diabetes and malnutrition remains limited [2].

Deep learning, convolutional neural networks and models like YOLO can recognize foods in photos pretty well.[1], [3] Deep learning is really good at this. YOLO models are good at recognizing foods in photos. When it comes to using this image based food recognition for personalized diet planning in one app that is easy to use and interactive is still not done. This is especially true for people in India. Learning and YOLO models are not being used to help people in india for personalized diet planning.

This paper is about the AI Meal Coach. The AI Meal Coach is a way to get nutrition advice. It does three things. First, it uses a custom YOLOv8 model that's specifically trained to recognize Indian dishes [5]. Second it uses a system with three parts to understand what the user wants to make a fitness plan and to create meals that fit the users needs [13]. Third it has a conversation interface that's safe to use. This interface is like a machine that can have a conversation with the user. The AI Meal Coach is really easy to use since you can take a picture of your food. Get personal diet advice.

Here's how the rest of the paper pans out: Section II walks through related research. Section III covers the system architecture. Section IV lays out the methodology, while Section V shows the implementation and working. Section VI shares evaluation results. The final sections wrap up with conclusions and what comes next.

II. LITERATURE SURVEY

A. Existing Systems

Many studies have been done to explore automated food recognition and nutrition tracking in the context of personal dietary management.

1. **CNN-Based Food Image Recognition Systems:** Early Deep learning systems used the CNN approach to classify food items from images [1], [3]. While it was good for Western food datasets, these systems lacked portion estimation, calorie calculation, and support for regional cuisines like Indian food.
2. **Mobile and Edge-Based Food Recognition Systems:** Deep learning based food recognition approach deployed on mobile and edge computing platforms has improved dietary assessment capabilities [2]. However, these systems do not have the capability of nutrition datasets integration, personalised dietary plan for each user, which results in limiting their practical utility.
3. **Calorie and Nutrition Estimation Frameworks:** Research systems have attempted to estimate nutritional value directly from food images using segmentation and depth-based volume estimation [7], [8]. These methods usually need conditions to work and they do not work with planning a diet for users or talking to a computer in a normal way. The methods do not really help with diet planning or conversational interfaces.
4. **AI-Driven Diet Recommendation Systems:** Machine learning based diet recommendation platforms use user health profiles to suggest meal plans [11]. However, these systems generally operate independently from food detection pipelines and lack real-time image-based intake logging or safety-validated outputs.

B. Related Research

1. Jiang et al. — DeepFood (2020) introduced the DeepFood framework, which integrates food image recognition with dietary assessment using deep neural networks [8]. While the system successfully combined visual analysis with nutritional evaluation, it did not incorporate personalized recommendation capabilities or interactive user engagement through conversational interfaces.

2. Chen et al. — Multi-Task Deep Learning (2021) proposed a multi-task deep learning framework capable of simultaneously identifying food items and their constituent ingredients from a single image [3]. This approach improved recognition efficiency by combining multiple tasks into a unified model. However, the system lacked mechanisms for calorie estimation and did not provide personalized dietary recommendations based on userspecific health parameters.

3. Mohanty et al. — The Food Recognition Benchmark (2022) established an open, reproducible benchmark for food recognition using the MyFoodRepo-273 dataset of 24,119 real-world food images spanning 273 categories [1]. Their work demonstrated reliable deep learning-based detection with top models achieving a mean average precision of 0.568, but the dataset focused primarily on

Swiss food items and lacked integration with nutritional estimation or personalized dietary recommendations.

4. Yang et al. — ChatDiet (2024) presented ChatDiet, a conversational diet recommendation system powered by large language models [9]. The system enables personalized dietary guidance through natural language interaction, enhancing user engagement. However, it does not include an image-based food recognition component, thereby relying entirely on user input for food logging.

C. Key Insights

From the literature, the following insights were identified:

- Existing food detection systems primarily focus on visual classification and object recognition without extending their functionality to comprehensive nutritional analysis. Many models are trained on datasets dominated by Western cuisine, resulting in poor generalization to region-specific foods, particularly Indian dishes.
- Diet recommendation systems typically operate independently of food detection pipelines. These systems rely heavily on manual user input for food logging, which introduces inaccuracies and reduces user engagement.
- Conversational diet chatbots have emerged as an interactive approach to personalized nutrition guidance, leveraging large language models to generate responses. While these systems enhance user engagement through natural language interaction, they are largely limited to text-based inputs and lack integration with computer vision models.
- The proposed system addresses these gaps by integrating food detection, nutrition estimation, personalized diet recommendation, and conversational interaction into a unified architecture. Unlike existing approaches, it incorporates a computer vision model optimized for Indian cuisine, enabling accurate recognition of region-specific foods.

A brief comparative study of selected existing systems was carried out to identify their key limitations and improvement areas. The major limitations and inferred future scope of these works are summarized in **Table 2.1**, as shown below.

Table 2.1 Limitation Of Existing Systems

Title	Limitation and Future Scope Inferred
Jiang et al. (2020): DeepFood — Food Image Analysis and Dietary Assessment via Deep Model	Jiang et al. presented DeepFood, a deep learning-based system that combined food image analysis with dietary assessment to evaluate nutritional content from visual inputs [8]. The

	framework demonstrated promising results in bridging computer vision with nutritional evaluation. However, the system did not incorporate personalized recommendation capabilities based on individual health profiles, nor did it provide any interactive user engagement through conversational interfaces, limiting its practical usability for daily dietary management.		regional contexts such as Indian food environments.
Chen et al. (2021): Multi-Task Deep Learning for Food Ingredient Recognition	Chen et al. proposed a multi-task deep learning framework capable of simultaneously recognizing food items and identifying their constituent ingredients from a single image [3]. This approach improved upon single-task classification models by extracting richer food-level semantic information. However, the system lacked mechanisms for calorie estimation and did not provide personalized dietary recommendations based on user-specific health parameters, restricting its utility as a complete nutrition management tool.	Yang et al. (2024): ChatDiet — LLM-Augmented Personalized Nutrition Chatbot	Yang et al. proposed ChatDiet, a conversational nutrition assistant powered by large language models that delivers personalized food recommendations through natural dialogue [9]. The system demonstrated strong contextual understanding and the ability to tailor dietary suggestions based on user preferences expressed in conversation. However, ChatDiet does not include an image-based food recognition component, thereby relying entirely on manual user input for food logging, which introduces the same friction and inaccuracy limitations found in traditional diet tracking applications.
Mohanty et al. (2022): The Food Recognition Benchmark — Using Deep Learning to Recognize Food in Images	Mohanty et al. developed the MyFoodRepo-273 benchmark dataset comprising 24,119 real-world food images across 273 categories, collected through a mobile health app and annotated by professional annotators [1]. Top-performing deep learning models achieved a mean average precision of 0.568. However, the dataset focused predominantly on Swiss food items and did not incorporate nutritional estimation or personalized dietary recommendations, limiting its applicability in diverse cultural and		

III. Proposed System

A. System Overview

The AI Meal Coach system is an intelligent, AI-powered nutrition estimation and personalized diet recommendation platform designed to help users understand their food intake and maintain a healthier lifestyle. In today's fast-paced environment, individuals often struggle to track their daily nutrition accurately, as manual calorie counting is time-consuming, error-prone, and lacks personalization. The proposed system addresses this challenge by automating food recognition and providing tailored dietary guidance based on individual user profiles.

The system allows users to upload or capture an image of their meal through the application interface. It then automatically detects food items using a computer vision model, estimates their nutritional content through an integrated data pipeline, and logs the information for daily tracking. In addition, users can interact with an AI-powered chatbot to receive personalized diet plans, fitness recommendations, and answers to nutrition-related queries. The platform also provides a dashboard where users can monitor their daily calorie intake, macronutrient distribution, and overall progress.

B. System Architecture Diagram

The system architecture of the proposed AI Meal Coach, an intelligent nutrition estimation and personalized diet recommendation platform, illustrates the overall structural design and workflow of the application. It demonstrates how multiple components—including the frontend interface, food detection module, machine learning pipeline, and data storage systems—interact to deliver an end-to-end user experience for automated nutrition tracking and dietary guidance.



Fig. 3.2. System Workflow of AI Meal Coach

C. Working of the Proposed System

The way the proposed AI Meal Coach system works is pretty straightforward. It follows an automatic process that you can see in the system architecture diagram.

First you use the application by taking a picture of your meal or uploading one through the Food Scanner part. The frontend gets this picture. Sends it to the food detection backend to be looked at.

The backend then uses a tool called YOLOv8 to figure out what food is in the picture. This tool looks at the picture. Finds the different food items and how many of each there are. The AI Meal Coach system then sends this information back, to the frontend in a way that's easy to understand so it can be used for other things. The AI Meal Coach system is what makes all of this work together.

When the system figures out what food items you have it starts to estimate the nutrition. First it tries to get the nutrition information from a database on your device. If it cannot find the information it asks a nutrition service for help. If that

does not work it uses a default value so the system keeps working. Then it shows you the nutrition values it calculated and asks you to confirm.

After you confirm the system saves the meal information and its nutrition details in the Firebase Realtime Database. This way the system can keep track of what you eat every day calculate how calories you take in and see how the different parts of your food like proteins and fats change over time.

At the time the system has a smart conversation interface. When you send a message through the chat the system sends your message to the chatbot server using an API. The server uses a set of rules and a three-part machine learning process to understand what you mean. The first part figures out what you want the second part makes a fitness plan with calorie goals and the third part creates a complete meal plan based on what you like and what you can eat.

Finally the system shows you the results on the dashboard and in the chat. Here you can see what you have eaten get a diet plan and track how your health is doing. The system works smoothly from start, to finish saving you time and giving you personalized diet advice in real time.

A. System Architecture

The proposed system integrates the following components:

- **Frontend:** React 18 Tailwind CSS for building a responsive, user-friendly, and interactive interface, including modules such as food scanning, dashboard visualization, meal logging, and AI chat..
- **Chatbot Backend:** Flask-based REST API responsible for handling user interactions, managing conversation flow through a state machine, and executing the three-model machine learning pipeline for personalized diet recommendations.
- **Food Detection Engine:** YOLOv8-based computer vision model deployed on a separate Flask server (Port 5001) for real-time food item detection from user-uploaded images.
- **Machine Learning Pipeline:** A three-stage pipeline consisting of an intent classifier (TF-IDF + classifier), a fitness roadmap generator (Random Forest), and a rule-based meal recommendation engine for generating personalized diet plans.
- **Safety Validation Module:** A rule-based validation system that enforces constraints such as allergen filtering, portion control, protein requirements, and calorie limits to ensure safe and reliable dietary recommendations.
- **Database & Authentication:** Firebase Realtime Database for secure storage of user profiles, meal logs, and daily nutrition data, along with Firebase Authentication for user management and access control.

B. Workflow

1. User opens the application and registers or logs in using Firebase Authentication.

2. User completes the onboarding process by providing personal details including age, weight, height, fitness goal, dietary preference, food restrictions, and activity level.
3. User navigates to the Food Scanner page and uploads or captures an image of their meal.
4. The image is sent to the Food Detection Backend where the YOLOv8 model detects food items and returns their labels and counts.
5. The frontend performs a cascading nutrition lookup — first checking the local Indian food database, then the USDA FoodData Central API, and finally applying a zero-default if no data is found.
6. User confirms the detected meal and nutritional data is stored in Firebase Realtime Database under the current date.
7. User opens the AI Chat interface and types a message or requests a diet plan.
8. The Chatbot Backend receives the message, processes it through the conversation state machine, and routes it through the three-model ML pipeline.
9. Model 1 classifies the user's intent, Model 2 generates a fitness roadmap with target calories, and Model 3 produces a complete daily meal plan.
10. The generated meal plan passes through the four-gate Safety Validator checking for allergens, portion sizes, protein minimums, and calorie tolerance.
11. The validated diet plan and recommendations are returned to the user through the AI Chat interface and reflected on the Dashboard.

IV. Methodology

The proposed AI Meal Coach – Intelligent Nutrition Estimation and Personalized Diet Recommendation System is designed as an integrated, user-centric platform that enables automated food recognition, nutritional analysis, and personalized dietary guidance through both visual and conversational inputs. The system combines computer vision, machine learning pipelines, and real-time data processing to deliver accurate and practical health insights. The overall workflow of the system is divided into five primary phases: Food Image Acquisition and Detection, Nutrition Estimation and Data Processing, Meal Logging and Storage, Conversational Recommendation Generation, and Safety Validation with User Interaction, as illustrated in Fig. 4.

A. User Authentication and Onboarding

The system starts with a way to log in which is controlled by Firebase Authentication. This makes sure that the information of the users is private and that only the right people can get to it. When people use the application they can sign up. Log in to see things that are just for them. After they log in they go to the page, which is where they can get to all the things the system can do.

The system keeps track of things that're specific to each user like what they like to eat what they want to achieve with their health what they are allergic, to and what they have eaten before. This information is stored in the Firebase Realtime

Database. The system uses this information to give users suggestions that're just for them. It also helps the system remember what the user did time they used it so it can make the experience better for them. The system does all of this to make sure the users have a time when they use it.

B. Food Input and Image Processing

When people log in they start using the Food Scanner part of the system. This is where they can take pictures of the food they eat or upload pictures from their phone. The part of the system that people see checks the picture to make sure it is okay sends it to the back end of the system using a special kind of message.

The back end of the system uses a tool called YOLOv8 to look at the picture of the food [5]. The Food Scanner module and the YOLOv8 tool work together to figure out what is in the picture like what kind of food it's how much of it there is. The Food Scanner module is important because it takes a picture of the food and turns it into information that the system can understand. This information is used to analyze the value of the food and the Food Scanner module is the starting point, for this analysis.

C. Nutrition Estimation and Meal Logging

The food items we find are sent to the nutrition estimation part of the system. This part gets the values by trying a few things one after the other. First it looks in a database on our system for answers. If it does not find what it needs it gets the information from places like the USDA FoodData database [14], and uses some standard values if it has to.

We then show the user the information we found and ask them to confirm it is correct. Once they say it is okay we store the details of the meal including how calories it has and what is, in it in the Firebase Realtime Database. This helps us keep track of what the user eats every day see what nutrients they are getting and help them understand what they are eating over time.

D. Conversational Recommendation and ML Pipeline

The system has a chatbot that people can talk to in their words. When people ask the chatbot something it sends the question to the backend, which figures out what to do using a special set of rules to keep the conversation going.

There are three steps that happen after that. The first stage performs intent classification using TF-IDF and machine learning models [12] to understand what the person really wants. The second stage generates a personalized fitness roadmap based on user goals using a Random Forest model [13]. The third stage produces a customized meal plan using a rule-based recommendation engine that considers nutritional requirements and user preferences.

E. Safety Validation and System Integration

The generated meal plans undergo a safety validation process to ensure reliability and health compliance. The system enforces multiple constraints, including allergen filtering, calorie limits, protein requirements, and balanced nutrient distribution through a multi-gate validation mechanism. Validated recommendations are then delivered to the user

through the dashboard and AI chat interface. The system integrates multiple components, including frontend modules, backend services, machine learning models, and external APIs such as Gemini and GROQ [15], ensuring seamless communication and efficient data processing.

V. RESULTS AND DISCUSSION

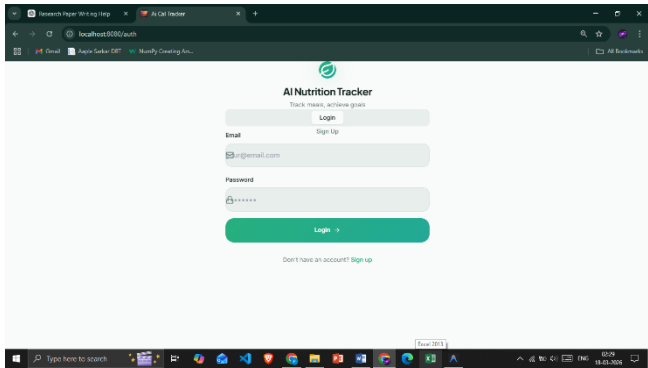


Fig 5.1.Login & Sign up Page

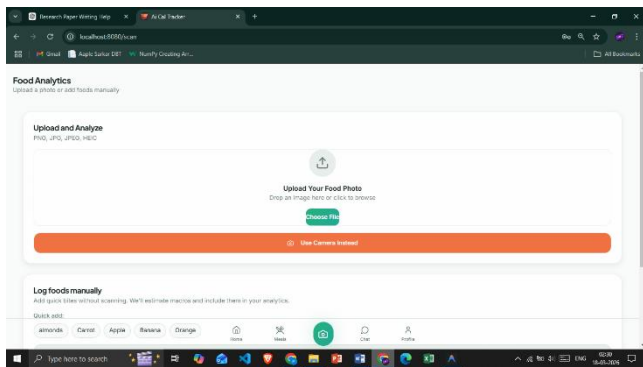


Fig 5.2. Food Scan Page

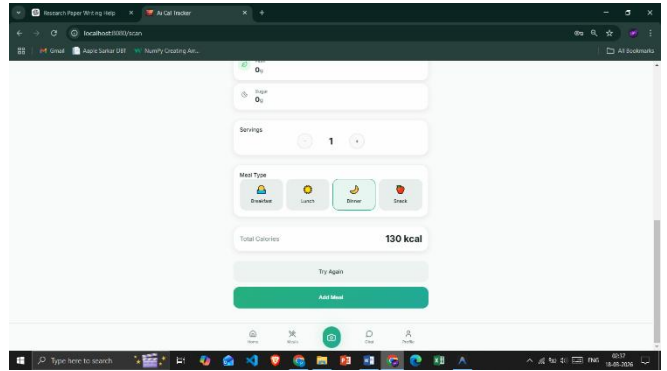
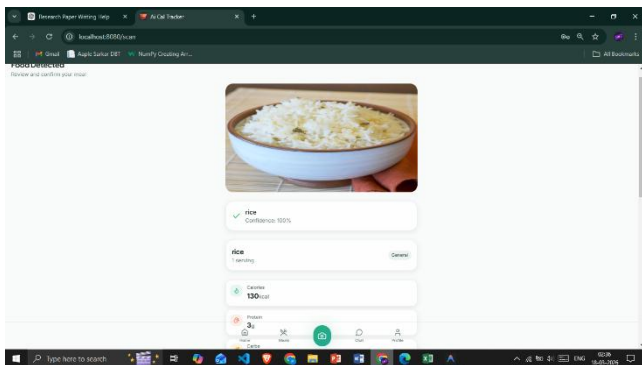


Fig 5.4. Scan Result 2

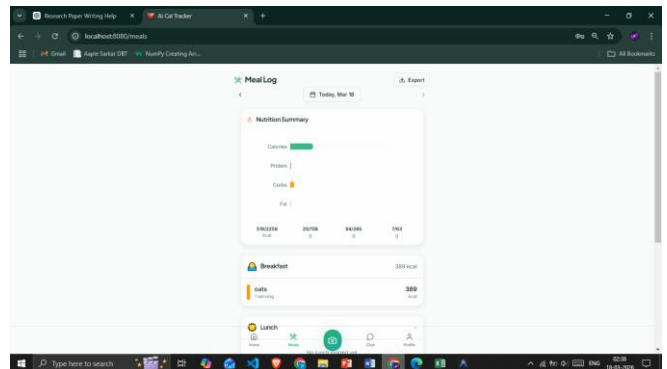


Fig 5.5. Meals Page

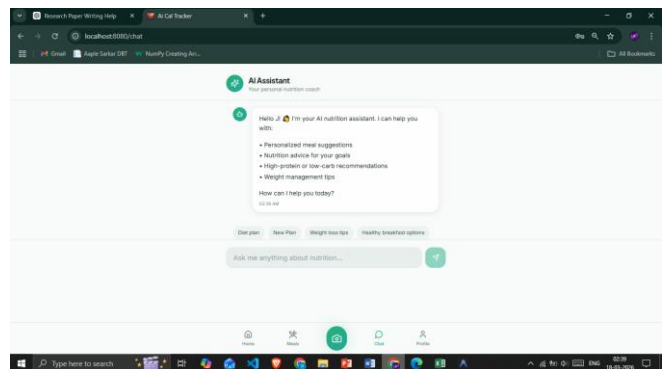


Fig 5.6. Chat page

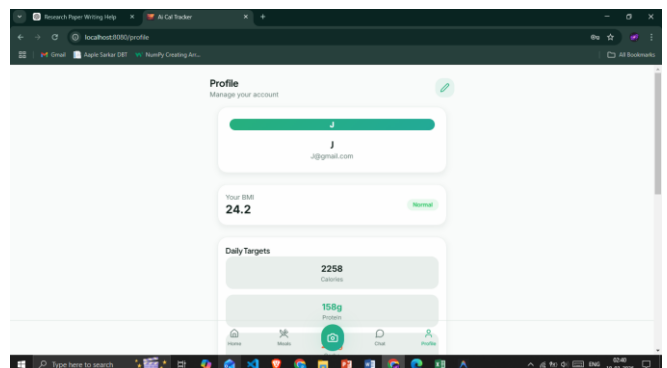


Fig 5.7. Profile Page

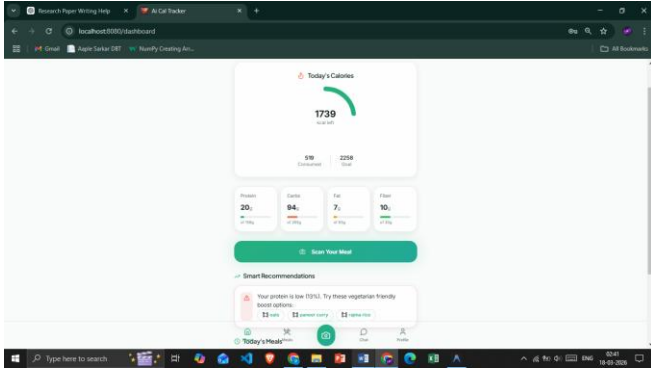


Fig 5.8. Home Page

The proposed AI Meal Coach system demonstrated reliable performance and accurate functionality during testing across multiple modules, including food detection, nutrition estimation, and personalized recommendation generation. The integration of automated food recognition and real-time nutritional analysis significantly improved user experience by reducing the manual effort required for calorie tracking and diet planning. The system consistently identified food items from user-uploaded images and generated structured nutritional outputs, enabling users to understand their dietary intake more effectively.

The implementation of the YOLOv8-based food detection model and the cascading nutrition lookup pipeline enabled efficient processing of real-world food images under varying conditions [5], [7]. The system successfully handled diverse food inputs, including region-specific dishes, and provided consistent nutritional estimations through a combination of local database queries and external API integration. Similar research highlights the importance of combining computer vision with structured nutritional databases to enhance accuracy and usability in dietary monitoring systems [7].

The conversational AI module further enhanced system interaction by enabling users to receive personalized diet recommendations through natural language queries. The three-stage machine learning pipeline—comprising intent classification, fitness roadmap generation, and meal plan recommendation—produced structured and goal-oriented outputs. The integration of a state machine ensured controlled and context-aware interactions, improving response relevance and user engagement.

Scalability testing indicated that the system could efficiently handle multiple concurrent users and simultaneous requests due to its modular architecture. The separation of frontend, food detection backend, and chatbot backend ensured distributed processing, reducing system load and improving response time. The use of Firebase Realtime Database enabled seamless data synchronization across modules, ensuring consistent updates of meal logs, nutritional data, and user profiles without latency issues.

Overall, the AI Meal Coach system achieved high reliability, efficient performance, and strong integration across all modules. The combination of computer vision, machine learning, and conversational AI within a unified platform provides a practical and scalable solution for intelligent dietary management. The results demonstrate that such integrated systems can significantly improve user

awareness, promote healthier eating habits, and enhance decision-making in everyday nutrition tracking scenarios.

VI. PERFORMANCE EVALUATION

To evaluate the effectiveness of the proposed AI Meal Coach – Intelligent Nutrition Estimation and Personalized Diet Recommendation System, a comprehensive performance analysis was conducted. The system was assessed based on specific parameters including detection model used, Indian cuisine support, food ingredient recognition accuracy, nutrition estimation method, personalized recommendation capability, conversational interface, safety validation, response time, training dataset, real-time processing, and database storage.

The summarized results of the evaluation are presented in Table 6.1, as shown below.

Table 6.1 Performance Evaluation of AI Meal Coach

Feature / Capability	AI Meal Coach (Proposed)	System 1: DeepFood [8]	System 2: ChatDiet [9]	System 3: IoMT Diet [11]
Food Image Detection	✓	✓	✗	✗
Indian Cuisine Support	✓	✗	Partial	✗
Nutrition Estimation	✓	Partial	✗	Partial
Personalized Diet Recommendation	✓	✗	✓	✓
Conversational Interface	✓	✗	✓	✗
Safety Validation	✓	✗	✗	✗
Real-time Processing	✓	Partial	✓	✓
Response Time	1.5–2.0 s	3–5 s	N/A	N/A
Database / Storage	✓	✗	✗	✓
Scalability (Multi-user)	✓	✗	Partial	✗

The performance of the AI Meal Coach system was evaluated based on functionality, response time, scalability, accuracy of food detection and nutrition estimation, and overall user interaction efficiency. The evaluation focused on assessing how effectively the system automates dietary tracking, generates personalized recommendations, and maintains data consistency across modules.

Testing was conducted across all major components, including food image processing, nutrition estimation, chatbot interaction, and database storage. The system

demonstrated stable and reliable performance during all test scenarios, with smooth coordination between the frontend interface, backend services, and machine learning modules.

The average response time for processing food images and generating nutritional output ranged between 2.0–3.5 seconds, depending on image complexity and network conditions. The YOLOv8-based detection model provided accurate results for a wide range of food items, including region-specific dishes, while the cascading nutrition lookup ensured consistent data availability [5].

Database operations managed through Firebase Realtime Database executed efficiently, enabling real-time updates of meal logs, calorie tracking, and user data without noticeable latency. The modular architecture and distributed processing approach allowed the system to maintain stable performance even under increased load.

Additionally, the safety validation module consistently enforced dietary constraints, ensuring that all generated meal plans were nutritionally balanced and safe for users. Overall, the system achieved high reliability, efficient performance, and effective integration, demonstrating its suitability as a scalable and intelligent dietary management solution.

VII. CONCLUSION

The proposed AI Meal Coach – Intelligent Nutrition Estimation and Personalized Diet Recommendation System successfully demonstrates the integration of computer vision, machine learning, and conversational AI to build a comprehensive and scalable dietary management solution. The system enables users to automatically detect food items from images, estimate nutritional values, track daily intake, and receive personalized diet recommendations through an interactive interface, thereby simplifying the process of maintaining a healthy lifestyle.

Future enhancements may include improving food detection accuracy using larger and more diverse datasets, incorporating portion size estimation through depth analysis, and enabling real-time video-based food tracking. Additional features such as wearable device integration, advanced health analytics, multilingual conversational support, and mobile application deployment can further expand system capabilities. Integration of predictive analytics for personalized diet planning and long-term health monitoring can also enhance user engagement and overall system impact.

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